Silencing as Input Rescaling: Predicting the Intensity-Dependent Effects of Neuron Loss from Connectome Structure

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1. Background and motivation

Complete synaptic wiring diagrams such as the FlyWire connectome of the adult *Drosophila* brain (Dorkenwald et al., 2024), together with spiking models built directly on them (Shiu et al., 2024), allow any neuron to be silenced in simulation and its behavioural consequence read out. A common working assumption is that neurons that are central in the graph are also functionally necessary. Experiments in the fly feeding circuit argue against a simple version of this view: silencing single second-order gustatory neurons often has modest or no effect on proboscis extension (Shiu et al., 2022; Jacobs et al., 2024). In the whole-brain model, Shiu et al. (2024) further observed that redundancy in the sugar pathway increases with stimulus strength, so that the effect of losing a neuron depends on how strongly the circuit is driven.

Two established frameworks describe closely related situations. In metabolic control analysis, control over pathway flux is shared among enzymes and the control coefficients sum to one, which explains why removing a single enzyme often barely alters flux (Kacser & Burns, 1981). In pharmacology, partial loss of a receptor reserve shifts the dose–response curve to the right before it lowers the maximal response. Neither framework has been equipped with an estimator derived from network structure, and neither has been tested in a signed, spiking, whole-brain connectome model. This project develops such an estimator and tests its predictions.

2. Research question and hypotheses

Can a single, structurally estimable quantity per neuron, the fraction f_k of sensory-to-motor transmission carried by neuron k, predict the full intensity-dependent consequence of silencing that neuron? Let $O(\lambda)$ denote the firing rate of the proboscis motor neuron MN9 when sugar gustatory receptor neurons (GRNs) are driven at rate λ, and $O_k(\lambda)$ the same rate with neuron k silenced.

H1. Classical centrality measures (degree, synaptic strength, PageRank, betweenness, k-core, output synapse count) predict single-neuron necessity poorly and inconsistently across stimulus intensities.

H2 (input-rescaling law). Silencing acts as a weakening of the stimulus, $O_k(\lambda) = O\big((1 - f_k)\,\lambda\big)$: the dose–response curve shifts by a factor $1/(1 - f_k)$ along the GRN-rate axis while its maximum is preserved. A neuron that is a source-to-target vertex cut should abolish the response at every tested intensity; the converse is not expected to hold in a threshold network.

H3. f_k can be estimated without simulating any perturbation, from walk-sum (linear-response) and max-flow fractions computed on the connectome and weighted by intact activity, and this estimate outperforms both classical centralities and the neuron's baseline firing rate.

H4 (pilot). The response to silencing a pair of neurons is predicted by their individual fractions together with an independently computed measure of their shared routes.

H5. Across a set of neurons that separates the sensory input from the motor output, the fractions sum to approximately one, and the departure from one, which measures redundancy, changes systematically with stimulus intensity.

3. Methods

Model and stimulation. We use the leaky integrate-and-fire model of Shiu et al. (2024) on FlyWire v783 (138,639 neurons), with v630 as a replication. Twenty right-hemisphere sugar GRNs receive Poisson input at rate λ. The primary readout is the MN9 contralateral to the stimulated GRNs; the ipsilateral MN9 is analysed separately. Simulations use an optimized re-implementation of the model that reproduces the original simulator spike for spike under identical input and is roughly two orders of magnitude faster.

Silencing experiments. An intact dose–response curve (0–400 Hz in 5 Hz steps, 60 trials) defines a 10-level, logarithmically spaced intensity grid spanning 0.5 to 8 times the half-maximal intensity, fixed before any silencing simulation. Each of the 573 neurons active under sugar drive is silenced individually across this grid (30 trials, with common random numbers so that silenced and intact conditions can be compared trial by trial). The design also includes positive and negative controls, whole-cell-type silencing for external validation, a stratified pilot of 60 neuron pairs, and a recruitment control in which random halves of the input population are driven.

Curve analysis. Each silenced dose–response curve is fitted with three nested descriptions: a pure horizontal shift (H2), a shift with an additional gain change, and an unconstrained Hill function. Model choice is made by cross-validated prediction error, and the shift model is additionally checked with an absolute goodness-of-fit test on held-out trials. Neurons whose curves cannot distinguish between the models are reported as uninformative rather than counted as support.

Structural predictors. All predictors are computed before any silencing result is examined. The baseline family comprises the classical centralities above and the hop distance to MN9. The structure-only family comprises the walk-sum fraction (computed exactly with Sherman–Morrison updates), the excitatory max-flow fraction and dominator status. The activity-aware family weights the walk-sum and max-flow fractions by neuronal gains measured on the intact dose–response curves.

External validation. For ten cell types with published optogenetic silencing data at two sucrose concentrations (Shiu et al., 2022), behavioural outcomes were assigned to three tiers in advance of any model analysis (Rattle and Usnea strongest; Zorro, Clavicle, G2N-1 and FMIn intermediate; Phantom without effect), with Bract, Fdg and Roundup forming a downstream layer without effect. Model fractions are obtained by silencing all members of each type together.

4. Statistical analysis

The analysis plan was fixed before any silencing simulation. The primary test compares rank correlations with the simulated fractions $\widehat{f}$: $\Delta\rho = \rho_s(f^{\text{LIN-A}}, \widehat{f}) - \max_X \rho_s(X, \widehat{f})$, where $f^{\text{LIN-A}}$ is the activity-aware walk-sum fraction and X ranges over the classical centralities. Confidence intervals come from a bootstrap that resamples cell types (about 200 effective units, 2,000 resamples) and re-selects the best competitor in every resample; H3 is supported if the one-sided 95% lower bound exceeds zero. Five secondary tests, corrected with the Holm procedure, compare the activity-aware estimator with baseline firing rate, the structure-only estimator with centrality, the outcome of dominator neurons (exact binomial test), the agreement between type-level fractions and the experimental tiers (Kendall's τ_b with an exact permutation null), and the separation between second-order and downstream types (exact Wilcoxon test). The fraction of neurons obeying H2, the summation test (H5), the pair pilot (H4) and the replications (v630, left-hemisphere input, $\pm 20\%$ synaptic weight, and the original authors' published silencing simulations) are reported as estimates with confidence intervals.

5. Expected results and falsification

We expect most neurons with informative curves to follow the rescaling law; the claim will be rejected if the upper 95% confidence bound on this fraction lies below one half. We expect the activity-aware flow fraction to rank neurons more accurately than any centrality measure, and loss of response across all intensities to be concentrated in vertex-cut neurons. Deviations are anticipated for inhibitory neurons whose influence depends on stimulus intensity, for parallel routes with different thresholds, and for transmission that relies on coincident input near threshold. Where the law fails, the pattern of failures identifies the conditions under which connectome structure is insufficient to predict function.

6. Limitations

All conclusions concern the model rather than the animal; only the external comparison, which involves ten cell types, bears directly on biology. The quantity f_k is a finite-change analogue of a flux control coefficient, so the contribution lies in its structural estimation and its dose–response consequences rather than in the concept itself. The model omits neuromodulation and electrical synapses and is known to misrepresent some GABAergic second-order neurons, including Phantom (Shiu et al., 2024); disagreement confined to Usnea and Phantom will therefore be attributed to transmitter sign rather than to the law. Finally, because the model and the predictors share the same wiring diagram, part of any structure–function agreement is expected by construction; the substantive question is which compressions of the connectome are sufficient.

References

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